CLAIMS

1. A method for evaluating the health risk of selenium (Se) deficiency in a subject comprising

a) determining in a sample of said subject a measured biological value selected from a1) selenium (Se) and/or a2) selenoprotein P (SELENOP); and

b) determining in a sample of said subject a measured biological value of antibodies against SELENOP (SELENOP-Ab).

2. The method for evaluating the health risk of selenium (Se) deficiency in a subject according to claim 1 further comprising the steps of

c) setting two or more biological parameters into correlation, wherein said biological parameters are represented by the said measured biological values obtainable as defined in steps a) and b) of claim 1;

d) calculating a risk index score for the Se deficiency of said subject using the values obtainable as defined i) in step a) and b) of claim 1 or ii) in step c).

3. The method according to claim 1 or 2, wherein the said measured biological values are correlated with the risk related to Selenium deficiency in said subject, wherein said subject is suffering from a disorder selected from sepsis, inflammation, SIRS, inflammatory bowel diseases, infection, cardio-vascular diseases (CVD), atherosclerosis, cardiovascular event, stroke, myocardial infarction, cardiomyopathy, cancer, breast cancer, prostate cancer, colorectal cancer, lung cancer, liver cancer, (poly)trauma, thyroid dysfunction, diabetes mellitus type I and/or II, a severe illness treated in an intensive care unit (ICU), severe burn or scald injury, poisoning, a neurodegenerative disease, epilepsia, Alzheimer's disease, Parkinson's disease, Huntington's disease, progressive or chronical multiple sclerosis (MS), amyotrophic lateral sclerosis (ALS), infertility, or hypertension.

4. The method according to one or more of claims 1 to 3, wherein step a2 and/or step b is performed as an immunoassay.

5. The method according to one or more of claims 1 to 3, which includes a spectrometric method selected from mass spectrometry, fluorescence spectrometry, X-ray fluorescence spectroscopy, and absorption spectrometry.

6. The use of the method as defined in one or more of claims 1 to 5 or the risk index score obtainable according to one or more of claims 2 to 5 as a companion diagnostic method to monitor Se supplementation in a subject.

7. The use of the method as defined in one or more of claims 1 to 5 or the risk index score obtainable according to one or more of claims 2 to 5 in the quantification of the individual Se requirement and dosage regime during therapeutic or prophylactic Se supplementation in a subject.

8. The use of the method as defined in one or more of claims 1 to 5 or the risk index score obtainable according to one or more of claims 2 to 5 in the analysis and monitoring of Se poisoning or the treatment of a subject suffering from selenosis by administration of a pharmaceutically effective amount of an antibody against SELENOP.

9. The use of the method as defined in one or more of claims 1 to 10 or the risk index score obtainable according to one or more of claims 1 to 10 in the treatment of a subject suffering from a disease selected from sepsis, infection, (poly)trauma or SIRS.

10. The use of the method as defined in one or more of claims 1 to 10 or the risk index score obtainable according to one or more of claims 1 to 10 in the treatment of a subject suffering from a disease selected from viral infection including influenza A and COVID-19.

11. A kit for performing the method as defined in claim 1 comprising
a) a specific reagent for the determination of antibodies against SELENOP (SELENOP-Ab),
b) a user leaflet;

c) optionally one or more reagents or chemicals selected from buffer, calibration standard, negative control, positive control; and

d) optionally a specific reagent for the determination of SELENOP.